Danmarks Tekniske Universitet

MODELING CO2 CONCENTRATIONS USING HIDDEN MARKOV MODELS

2026

Special Course in Statistical Modelling

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6. september 2026

Model	#Params	LogLik	AIC	BIC
HMM(1)	20	-10388.87	20815.73	20918.68
AR(1), HMM(1)	24	-9359.73	18765.45	18890.06
AR(2), HMM(1)	28	-9273.19	18600.38	18746.64
AR(1), HMM(2)	84	-9130.90	18427.80	18877.46
AR(2), HMM(2)	100	-9045.40	18288.79	18825.07
Covariates-HMM(1)	44	-10229.71	20545.41	20778.39
AR(1), Covariates-HMM(1)	48	-9272.25	18638.51	18893.13
AR(2), Covariates-HMM(1)	52	-9192.95	18487.90	18764.16

Tabel 1: Per-model log-likelihood, AIC and BIC for the fitted HMMs.

Base Model	Expanded Model	LRT	df	p-value	ΔAIC	ΔBIC
HMM(1)	AR(1), HMM(1)	2058.28	5	0.0000	-2050.28	-2028.61
AR(1), HMM(1)	AR(2), HMM(1)	173.07	5	0.0000	-165.07	-143.40
AR(1), HMM(1)	AR(1), HMM(2)	457.65	61	0.0000	-337.65	-12.60
AR(2), HMM(1)	AR(2), HMM(2)	455.59	73	0.0000	-311.59	78.43
AR(1), HMM(2)	AR(2), HMM(2)	171.01	17	0.0000	-139.01	-52.34
HMM(1)	Covariates-HMM(1)	318.32	25	0.0000	-270.32	-140.29
Covariates-HMM(1)	AR(1), Covariates-HMM(1)	1914.90	5	0.0000	-1906.90	-1885.23
AR(1), Covariates-HMM(1)	AR(2), Covariates-HMM(1)	158.61	5	0.0000	-150.61	-128.94
AR(1), HMM(1)	AR(1), Covariates-HMM(1)	174.95	25	0.0000	-126.95	3.08
AR(2), HMM(1)	AR(2), Covariates-HMM(1)	160.49	25	0.0000	-112.49	17.52

Tabel 2: Likelihood-ratio tests for each nested pair from the model hierarchy (Figure ??). Negative ΔAIC/ΔBIC favour the expanded model.

State	Mean ($\hat{\mu}$)	Std. Dev. ($\hat{\sigma}$)
State 1	400.000	21.748
State 2	535.684	76.573
State 3	889.928	110.981
State 4	1306.484	197.410

Tabel 3: Estimated emission parameters (Gaussian) for the ordinary 4-state HMM.

	State 1	State 2	State 3	State 4
State 1	0.8496	0.1476	0.0029	0.0000
State 2	0.0828	0.8486	0.0686	0.0000
State 3	0.0004	0.1605	0.6841	0.1550
State 4	0.0000	0.0000	0.0718	0.9282

Tabel 4: Estimated transition matrix for the ordinary 4-state HMM.

State	Mean ($\hat{\mu}$)	Std. Dev. ($\hat{\sigma}$)	AR coeff. ($\hat{\phi}$)
State 1	400.000	12.501	0.6235
State 2	509.326	77.704	0.3659
State 3	560.285	11.006	0.9462
State 4	1366.661	102.836	0.8318

Tabel 5: Estimated emission parameters for the AR-HMM (Gaussian with AR(1) residuals).

s_{t-1}	s_t	$\hat{\mu}$	$\hat{\sigma}$	$\hat{\phi}$
1	1	400.000	10.009	0.6797
2	1	400.000	12.472	0.2649
3	1	400.000	5.819	0.9871
4	1	400.000	81.272	1.0000
1	2	446.913	30.971	−0.0550
2	2	446.913	21.726	0.3599
3	2	446.913	32.836	0.1635
4	2	446.913	122.782	0.8514
1	3	574.844	12.245	0.9592
2	3	574.844	114.090	0.6235
3	3	574.844	0.001	0.8921
4	3	574.844	69.077	0.3838
1	4	1461.574	119.463	0.9186
2	4	1461.574	86.911	0.8508
3	4	1461.574	67.035	1.0000
4	4	1461.574	54.907	0.7415

Tabel 9: Estimated emission parameters for the AR(1) second-order HMM. Each row is a state pair (s_{t-1}, s_t); $\hat{\mu}$ and $\hat{\sigma}$ are the Gaussian mean and standard deviation of the residual and $\hat{\phi}$ is the AR(1) coefficient.

	State 1	State 2	State 3	State 4
State 1	0.7521	0.1685	0.0016	0.0779
State 2	0.2698	0.5960	0.0633	0.0709
State 3	0.0516	0.1027	0.8457	0.0000
State 4	0.0067	0.0492	0.0153	0.9289

Tabel 6: Estimated transition matrix for the AR-HMM.

State	Mean ($\hat{\mu}$)	Std. Dev. ($\hat{\sigma}$)	AR coeff. ($\hat{\phi}_1$)	AR coeff. ($\hat{\phi}_2$)	$\hat{\phi}_1 + \hat{\phi}_2$
State 1	400.000	10.765	0.7027	-0.0781	0.6247
State 2	525.630	72.083	0.5941	-0.2109	0.3832
State 3	716.015	16.158	0.9596	0.0565	1.0161
State 4	1299.660	97.603	0.9997	-0.2093	0.7904

Tabel 7: Estimated emission parameters for the AR(2)-HMM (Gaussian with AR(2) residuals).

From (s_{t-1}, s_t)	$s_{t+1} = 1$	$s_{t+1} = 2$	$s_{t+1} = 3$	$s_{t+1} = 4$
$(1, 1)$	0.7633	0.1270	0.0208	0.0889
$(1, 2)$	0.0705	0.9294	–	–
$(1, 3)$	0.7629	–	0.0001	0.2370
$(1, 4)$	0.7065	0.0663	0.1706	0.0565
$(2, 1)$	0.6903	0.0497	–	0.2600
$(2, 2)$	0.1675	0.6206	0.0153	0.1966
$(2, 3)$	–	0.0738	–	0.9262
$(2, 4)$	0.0242	0.1217	–	0.8542
$(3, 1)$	0.1370	–	0.8630	–
$(3, 2)$	0.5383	0.3560	0.1056	0.0001
$(3, 3)$	0.0425	–	0.0010	0.9564
$(3, 4)$	0.5295	0.1390	0.3313	0.0001
$(4, 1)$	0.1148	–	0.3133	0.5719
$(4, 2)$	0.0124	0.0767	0.3351	0.5757
$(4, 3)$	–	0.9998	–	0.0002
$(4, 4)$	–	0.1688	0.0135	0.8177

Tabel 10: Estimated transition probabilities for the AR(1) second-order HMM. Each row is a state pair (s_{t-1}, s_t); columns give the next-state probability $P(s_{t+1} \mid s_{t-1}, s_t)$. A dash (–) indicates zero or negligible probability.

	State 1	State 2	State 3	State 4
State 1	0.7115	0.2065	0.0000	0.0820
State 2	0.2848	0.5479	0.1055	0.0618
State 3	0.0216	0.1325	0.8459	0.0000
State 4	0.0061	0.0453	0.0171	0.9315

Tabel 8: Estimated transition matrix for the AR(2)-HMM.

s_{t-1}	s_t	$\hat{\mu}$	$\hat{\sigma}$	$\hat{\phi}_1$	$\hat{\phi}_2$	$\hat{\phi}_1 + \hat{\phi}_2$
1	1	400.000	7.818	0.7825	−0.0960	0.6865
2	1	400.000	12.809	0.4320	−0.1232	0.3088
3	1	400.000	6.407	1.0000	−0.0030	0.9970
4	1	400.000	126.469	1.0000	0.0524	1.0524
1	2	437.521	38.283	−0.2797	0.1697	−0.1100
2	2	437.521	16.359	0.5665	−0.0767	0.4897
3	2	437.521	67.879	0.7221	−0.1645	0.5576
4	2	437.521	113.350	1.0000	−0.1702	0.8298
1	3	570.631	6.012	0.9285	−0.0006	0.9279
2	3	570.631	98.402	−0.1690	0.4549	0.2859
3	3	570.631	0.000	0.9594	0.0327	0.9922
4	3	570.631	31.589	0.9997	0.0469	1.0466
1	4	1436.337	174.176	1.0000	−0.1229	0.8771
2	4	1436.337	94.026	0.9204	−0.0704	0.8500
3	4	1436.337	40.134	0.9998	0.0017	1.0015
4	4	1436.337	56.351	0.9458	−0.1708	0.7750

Tabel 11: Estimated emission parameters for the AR(2) second-order HMM. $\hat{\sigma}$ and the AR(2) coefficients $\hat{\phi}_1, \hat{\phi}_2$ depend on (s_{t-1}, s_t). A sum $\hat{\phi}_1 + \hat{\phi}_2 > 1$ indicates a non-stationary AR process for that state pair.

From (s_{t-1}, s_t)	$s_{t+1} = 1$	$s_{t+1} = 2$	$s_{t+1} = 3$	$s_{t+1} = 4$
$(1, 1)$	0.7192	0.2242	–	0.0565
$(1, 2)$	0.2137	0.4312	0.3548	0.0002
$(1, 3)$	0.7218	–	–	0.2782
$(1, 4)$	0.8343	0.1020	–	0.0637
$(2, 1)$	0.6948	0.3049	–	0.0003
$(2, 2)$	0.1289	0.6793	0.0328	0.1590
$(2, 3)$	–	0.5243	–	0.4757
$(2, 4)$	–	0.1968	–	0.8032
$(3, 1)$	0.1582	–	0.8418	–
$(3, 2)$	0.3295	0.5295	0.1410	–
$(3, 3)$	0.0008	–	0.3577	0.6415
$(3, 4)$	0.0409	0.1158	0.8432	0.0001
$(4, 1)$	0.0620	–	0.3651	0.5729
$(4, 2)$	–	0.0154	0.3636	0.6210
$(4, 3)$	0.1742	0.2723	–	0.5534
$(4, 4)$	–	0.1599	0.0272	0.8129

Tabel 12: Estimated transition probabilities for the AR(2) second-order HMM. Each row is a state pair (s_{t-1}, s_t); columns give the next-state probability $P(s_{t+1} \mid s_{t-1}, s_t)$. A dash (–) indicates zero or negligible probability.

State	Mean ($\hat{\mu}$)	Std. Dev. ($\hat{\sigma}$)
State 1	400.000	25.438
State 2	545.477	78.799
State 3	893.197	101.264
State 4	1302.677	198.527

Tabel 13: Estimated emission parameters (Gaussian) for the covariate HMM.

	State 1	State 2	State 3	State 4
State 1	0.6725	0.3275	0.0000	0.0000
State 2	0.0961	0.8563	0.0476	0.0000
State 3	0.0000	0.1027	0.8213	0.0761
State 4	0.0000	0.0000	0.0000	1.0000

Tabel 14: Estimated baseline transition matrix for the covariate HMM. This is the covariate-free intercept; the time-of-day covariates modulate these probabilities at each step.

From	To	$\beta_{\cos}$	$\beta_{\sin}$
State 1	State 2	0.8186	2.4047
State 1	State 3	31.3649	3.1998
State 1	State 4	0.8771	4.2476
State 2	State 1	−0.1781	0.1902
State 2	State 3	1.9664	2.6533
State 2	State 4	0.3837	1.6636
State 3	State 1	−31.5003	−5.3664
State 3	State 2	−0.7677	0.6256
State 3	State 4	2.5595	1.1320
State 4	State 1	−1.2404	−5.8053
State 4	State 2	−11.2206	−32.3030
State 4	State 3	−9.0565	6.1797

Tabel 15: Estimated covariate coefficients β for the covariate HMM. Each row is an off-diagonal transition State $i \to$ State j; columns give the effect of the time-of-day covariates $x_{\cos}$ and $x_{\sin}$ on that transition's logit.

State	Mean ($\hat{\mu}$)	Std. Dev. ($\hat{\sigma}$)	AR coeff. ($\hat{\phi}$)
State 1	400.000	11.953	0.5824
State 2	559.087	134.009	0.5670
State 3	1265.869	27.399	1.0000
State 4	1434.102	93.334	0.7984

Tabel 16: Estimated emission parameters for the AR(1) covariate HMM.

	State 1	State 2	State 3	State 4
State 1	0.8701	0.1298	0.0000	0.0001
State 2	0.2148	0.6578	0.1274	0.0000
State 3	0.0000	0.2233	0.7184	0.0584
State 4	0.0000	0.0000	0.0001	0.9999

Tabel 17: Estimated baseline transition matrix for the AR(1) covariate HMM. This is the covariate-free intercept; the time-of-day covariates modulate these probabilities at each step.

From	To	$\beta_{\cos}$	$\beta_{\sin}$
State 1	State 2	0.3703	−0.0332
State 1	State 3	45.3479	−28.5064
State 1	State 4	9.6651	2.6075
State 2	State 1	1.5104	−1.5301
State 2	State 3	−0.9594	−0.5055
State 2	State 4	0.5709	1.6448
State 3	State 1	−68.9564	15.5543
State 3	State 2	−0.0344	1.5140
State 3	State 4	1.9453	3.0555
State 4	State 1	−1.9853	−5.2421
State 4	State 2	−33.0764	−75.9324
State 4	State 3	−5.8518	7.0234

Tabel 18: Estimated covariate coefficients β for the AR(1) covariate HMM. Each row is an off-diagonal transition State $i \to$ State j; columns give the effect of the time-of-day covariates $x_{\cos}$ and $x_{\sin}$ on that transition's logit.

State	Mean ($\hat{\mu}$)	Std. Dev. ($\hat{\sigma}$)	AR coeff. ($\hat{\phi}_1$)	AR coeff. ($\hat{\phi}_2$)	$\hat{\phi}_1 + \hat{\phi}_2$
State 1	400.000	12.146	0.6429	-0.0565	0.5864
State 2	598.483	122.566	0.8803	-0.3048	0.5755
State 3	598.532	29.811	1.0000	0.0357	1.0357
State 4	1339.266	91.999	1.0000	-0.2230	0.7770

Tabel 19: Estimated emission parameters for the AR(2) covariate HMM.

	State 1	State 2	State 3	State 4
State 1	0.7492	0.2508	0.0000	0.0001
State 2	0.2898	0.5376	0.1726	0.0000
State 3	0.0000	0.1239	0.8464	0.0297
State 4	0.0000	0.0000	0.1925	0.8075

Tabel 20: Estimated baseline transition matrix for the AR(2) covariate HMM. This is the covariate-free intercept; the time-of-day covariates modulate these probabilities at each step.

From	To	$\beta_{\cos}$	$\beta_{\sin}$
State 1	State 2	0.2301	0.8881
State 1	State 3	39.8615	-26.4016
State 1	State 4	9.5134	3.2646
State 2	State 1	1.2788	-0.9649
State 2	State 3	-0.3258	-0.2435
State 2	State 4	0.6595	1.6863
State 3	State 1	-76.9450	16.4783
State 3	State 2	-0.6978	0.7277
State 3	State 4	2.4012	2.0695
State 4	State 1	-2.3714	-5.0618
State 4	State 2	-34.4690	-77.9890
State 4	State 3	-1.9356	0.4192

Tabel 21: Estimated covariate coefficients β for the AR(2) covariate HMM. Each row is an off-diagonal transition State $i \to$ State j; columns give the effect of the time-of-day covariates $x_{\cos}$ and $x_{\sin}$ on that transition's logit.